

Machine Learning

Introduction to Statistics

1. **What is the branch of mathematics that deals with the collection, analysis, and interpretation of data?**

- a) Calculus
- b) Statistics
- c) Algebra
- d) Geometry

Answer: b) Statistics

2. **What is a population in statistics?**

- a) A group of selected data points
- b) The entire group of individuals or observations of interest
- c) A subset of a sample
- d) None of the above

Answer: b) The entire group of individuals or observations of interest

Descriptive Statistics: Measures of Central Tendency

3. **Which of the following is not a measure of central tendency?**

- a) Mean
- b) Median
- c) Mode
- d) Range

Answer: d) Range

4. **What is the median of the dataset [5, 2, 9, 3, 8]?**

- a) 2
- b) 3
- c) 5
- d) 8

Answer: c) 5

Measure of Dispersion

5. Which measure of dispersion is defined as the difference between the maximum and minimum values?

- a) Variance
- b) Standard Deviation
- c) Range
- d) IQR

Answer: c) Range

6. Variance measures the _____ of data points around the mean.

- a) Centrality
- b) Spread
- c) Frequency
- d) None of the above

Answer: b) Spread

Handling Outliers

7. What is the interquartile range (IQR)?

- a) $Q3 - Q1$
- b) $Q3 + Q1$
- c) $Q2 - Q1$
- d) $Q1 + Q2$

Answer: a) $Q3 - Q1$

8. A data point is considered an outlier if it lies beyond _____ times the IQR from the quartiles.

- a) 1.0
- b) 1.5
- c) 2.0
- d) 2.5

Answer: b) 1.5

Measure of Symmetry (Skewness and Kurtosis)

9. What does a positive skew in a dataset indicate?

- a) A symmetric distribution
- b) A long tail on the right
- c) A long tail on the left
- d) None of the above

Answer: b) A long tail on the right

10. Kurtosis measures the _____ of a distribution.

- a) Central tendency
- b) Shape of tails
- c) Median
- d) Variance

Answer: b) Shape of tails

Inferential Statistics

Probability & Probability Distributions

11. What is the probability of an event that is certain to happen?

- a) 0
- b) 0.5
- c) 1
- d) None of the above

Answer: c) 1

12. Which of the following distributions is used for modeling the number of events in a fixed interval of time or space?

- a) Normal distribution
- b) Poisson distribution
- c) Binomial distribution
- d) Uniform distribution

Answer: b) Poisson distribution

Hypothesis Testing

13. What is the null hypothesis (H_0)?

- a) The statement to be proven true
- b) The statement to be tested
- c) A statement of no effect or no difference
- d) None of the above

Answer: c) A statement of no effect or no difference

14. Type I error occurs when:

- a) The null hypothesis is accepted when it is false
- b) The null hypothesis is rejected when it is true
- c) The alternate hypothesis is accepted when it is true
- d) None of the above

Answer: b) The null hypothesis is rejected when it is true

Tests (z-test, t-test, Chi-Square, F-Test, ANOVA, A/B Testing)

15. Which test is used to compare means of two groups when sample size is small?

- a) Z-test
- b) T-test
- c) F-test
- d) Chi-Square test

Answer: b) T-test

16. Chi-Square test is used to measure the relationship between:

- a) Two categorical variables
- b) Two numerical variables
- c) A categorical and a numerical variable
- d) None of the above

Answer: a) Two categorical variables

Preprocessing and Feature Engineering

Handling Missing Values

17. Which of the following is not a method to handle missing data?

- a) Mean imputation
- b) Median imputation
- c) Random initialization
- d) Deletion

Answer: c) Random initialization

Handling Outliers

18. Which method is commonly used to detect outliers in data?

- a) Boxplot
- b) Scatterplot
- c) Heatmap
- d) Histogram

Answer: a) Boxplot

19. What does the z-score indicate in outlier detection?

- a) The number of standard deviations a data point is from the mean
- b) The median value of the dataset
- c) The range of the dataset
- d) The minimum value of the dataset

Answer: a) The number of standard deviations a data point is from the mean

Treating Skewed Data

20. Which transformation is commonly used to reduce right skewness in data?

- a) Square root transformation
- b) Exponential transformation
- c) Log transformation
- d) Reciprocal transformation

Answer: c) Log transformation

21. What does a square root transformation do to the data?

- a) Reduces the skewness of large values
- b) Amplifies the skewness of small values
- c) Centers the data
- d) Normalizes the data

Answer: a) Reduces the skewness of large values

Feature Engineering

Label & One-hot Encoding

22. Label encoding is best used for:

- a) Ordinal categorical data
- b) Nominal categorical data
- c) Numerical data
- d) Continuous data

Answer: a) Ordinal categorical data

23. One-hot encoding is used to convert:

- a) Ordinal data into numerical data
- b) Nominal categorical data into binary features
- c) Continuous data into categorical data
- d) None of the above

Answer: b) Nominal categorical data into binary features

Feature Scaling

24. Which of the following is not a feature scaling method?

- a) Normalization
- b) Standardization
- c) One-hot encoding
- d) Min-max scaling

Answer: c) One-hot encoding

25. Standardization rescales data to have a mean of _____ and a standard deviation of _____.

- a) 0, 1
- b) 1, 0

c) 0, 0

d) 1, 1

Answer: a) 0, 1

Introduction to Artificial Intelligence and Machine Learning

Machine Learning Workflow

26. What is the purpose of the train_test_split function?

a) To combine datasets

b) To split the data into training and testing sets

c) To standardize the data

d) To encode categorical variables

Answer: b) To split the data into training and testing sets

27. What is a common split ratio for train-test datasets?

a) 70:30

b) 80:20

c) 90:10

d) All of the above

Answer: d) All of the above

Supervised Learning

Linear Regression

28. What does the coefficient of a feature represent in linear regression?

a) The probability of the feature

b) The mean of the feature

c) The change in the target variable for one unit increase in the feature

d) The standard deviation of the feature

Answer: c) The change in the target variable for one unit increase in the feature

29. Linear regression assumes a _____ relationship between input and output variables.

a) Linear

b) Non-linear

c) Quadratic

d) Exponential

Answer: a) Linear

Logistic Regression

30. What type of problem does logistic regression solve?

- a) Regression problems
- b) Clustering problems
- c) Classification problems
- d) Dimensionality reduction

Answer: c) Classification problems

31. Which function does logistic regression use to map predictions?

- a) Linear function
- b) Sigmoid function
- c) Quadratic function
- d) Exponential function

Answer: b) Sigmoid function

Decision Tree

32. What does a decision tree split data based on?

- a) Standard deviation
- b) Entropy or Gini index
- c) Mean value
- d) Range

Answer: b) Entropy or Gini index

33. Which of the following is a disadvantage of decision trees?

- a) Easy to interpret
- b) Prone to overfitting
- c) Handles non-linear data
- d) Fast training

Answer: b) Prone to overfitting

Ensemble Learning

34. What is the main idea behind ensemble learning?

- a) Use one strong model
- b) Combine multiple weak models
- c) Select the best single model
- d) Ignore weak models

Answer: b) Combine multiple weak models

35. Which of the following is an example of boosting?

- a) Random Forest
- b) Gradient Boosting
- c) KNN
- d) Decision Tree

Answer: b) Gradient Boosting

Model Evaluation

36. What is the F1 score a measure of?

- a) Precision and recall
- b) Accuracy and precision
- c) Sensitivity and specificity
- d) Recall and specificity

Answer: a) Precision and recall

37. Cross-validation is used to:

- a) Split data into training and testing sets
- b) Test model stability
- c) Select hyperparameters
- d) Train multiple models

Answer: b) Test model stability

Unsupervised Learning

Clustering

38. Which method is used to determine the optimal number of clusters in KMeans?

- a) Dendrogram

- b) Elbow method
 - c) Correlation matrix
 - d) Silhouette score
- Answer:** b) Elbow method

39. Agglomerative clustering is an example of:

- a) Partition-based clustering
 - b) Density-based clustering
 - c) Hierarchical clustering
 - d) KMeans clustering
- Answer:** c) Hierarchical clustering
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Dimensionality Reduction

40. Which algorithm is commonly used for dimensionality reduction?

- a) PCA
- b) KMeans
- c) Decision Tree
- d) Logistic Regression

Answer: a) PCA

41. Dimensionality reduction helps to:

- a) Increase computational complexity
- b) Reduce overfitting
- c) Increase the number of features
- d) None of the above

Answer: b) Reduce overfitting

Deep Learning

Neural Networks

42. What is the basic building block of a neural network?

- a) Node
- b) Weight
- c) Neuron
- d) Layer

Answer: c) Neuron

43. Which of the following is not an activation function?

- a) Sigmoid
- b) ReLU
- c) Tanh
- d) Mean Squared Error

Answer: d) Mean Squared Error

Supervised Learning (continued)

K-Nearest Neighbors (KNN)

44. What does K represent in the KNN algorithm?

- a) Number of features
- b) Number of neighbors to consider
- c) Number of iterations
- d) None of the above

Answer: b) Number of neighbors to consider

45. KNN is a _____ algorithm.

- a) Parametric
- b) Non-parametric
- c) Clustering
- d) Dimensionality reduction

Answer: b) Non-parametric

Support Vector Machines (SVM)

46. What does SVM aim to maximize?

- a) Distance between data points
- b) Margin between classes
- c) Sum of squared errors
- d) Entropy

Answer: b) Margin between classes

47. Which kernel in SVM is commonly used for non-linear data?

- a) Linear kernel
- b) Polynomial kernel
- c) RBF (Radial Basis Function) kernel

d) Both b and c

Answer: d) Both b and c

Naive Bayes

48. Naive Bayes is based on which theorem?

- a) Central Limit Theorem
- b) Bayes' Theorem
- c) Pythagoras' Theorem
- d) None of the above

Answer: b) Bayes' Theorem

49. Naive Bayes assumes features are:

- a) Dependent
- b) Independent
- c) Correlated
- d) None of the above

Answer: b) Independent

Regression & Classification Model Evaluation

50. Which metric is used to evaluate regression models?

- a) Accuracy
- b) R-squared
- c) Recall
- d) F1 Score

Answer: b) R-squared

51. What is a confusion matrix used for?

- a) Clustering evaluation
- b) Evaluating classification models
- c) Dimensionality reduction
- d) Preprocessing

Answer: b) Evaluating classification models

52. Which metric is appropriate for imbalanced classification problems?

- a) Accuracy
- b) Precision
- c) F1 Score
- d) Recall

Answer: c) F1 Score

53. Cross-validation helps to address which issue?

- a) Underfitting
- b) Overfitting
- c) Data imputation
- d) Data scaling

Answer: b) Overfitting

Unsupervised Learning (continued)

Clustering

54. What is the Random Initialization Trap in KMeans?

- a) Incorrect cluster assignment due to random initialization
- b) Convergence to optimal solution
- c) Use of hierarchical clustering
- d) None of the above

Answer: a) Incorrect cluster assignment due to random initialization

55. What is the purpose of a dendrogram in hierarchical clustering?

- a) To determine the optimal number of clusters
- b) To visualize feature importance
- c) To normalize data
- d) None of the above

Answer: a) To determine the optimal number of clusters

Dimensionality Reduction

56. Principal Component Analysis (PCA) reduces dimensions by focusing on:

- a) Variance
- b) Mean

c) Median

d) Range

Answer: a) Variance

57. What happens when too many dimensions are reduced using PCA?

a) Overfitting

b) Loss of information

c) Improved model accuracy

d) None of the above

Answer: b) Loss of information

Deep Learning

Neural Networks

58. What is the primary function of an activation function in neural networks?

a) To update weights

b) To add non-linearity to the model

c) To optimize loss

d) To normalize data

Answer: b) To add non-linearity to the model

59. Which of the following is a commonly used loss function in regression problems?

a) Cross-entropy loss

b) Mean Squared Error (MSE)

c) Binary Cross Entropy

d) Hinge Loss

Answer: b) Mean Squared Error (MSE)

60. What does backpropagation do in a neural network?

a) Propagates input data forward

b) Optimizes weights using gradients

c) Adds activation functions

d) Combines multiple layers

Answer: b) Optimizes weights using gradients

Handling Missing Values

61. Which imputation method is suitable for categorical variables?

- a) Mean imputation
- b) Median imputation
- c) Mode imputation
- d) Standardization

Answer: c) Mode imputation

62. Dropping rows with missing data is suitable when:

- a) The missing data percentage is low
- b) The missing data percentage is high
- c) The dataset is large
- d) The missing data is in numerical columns

Answer: a) The missing data percentage is low

Handling Duplication

63. What is the first step to handling duplicate records?

- a) Identify duplicates using unique IDs or columns
- b) Randomly delete records
- c) Normalize the data
- d) Perform feature scaling

Answer: a) Identify duplicates using unique IDs or columns

64. Which Python library is commonly used to handle duplicates in datasets?

- a) NumPy
- b) Pandas
- c) Scikit-learn
- d) TensorFlow

Answer: b) Pandas

Feature Selection

65. Which method is used to rank features by importance?

- a) SelectKBest
- b) Min-max scaling

c) Standardization

d) Normalization

Answer: a) SelectKBest

66. Correlation-based feature selection is suitable for:

a) Categorical features

b) Numerical features

c) Both categorical and numerical features

d) None of the above

Answer: b) Numerical features

67. Which statistical measure is often used to evaluate feature correlation?

a) Pearson's coefficient

b) F1 Score

c) Entropy

d) Gini index

Answer: a) Pearson's coefficient

A/B Testing

68. What is the primary goal of A/B testing?

a) To compare two versions of a variable

b) To create a new model

c) To reduce variance in data

d) None of the above

Answer: a) To compare two versions of a variable

69. In A/B testing, the control group receives:

a) The new variant

b) No treatment or the standard version

c) A combination of all variants

d) None of the above

Answer: b) No treatment or the standard version

Handling Outliers (continued)

70. The interquartile range (IQR) is defined as:

- a) $Q3 - Q1$
- b) $Q2 - Q1$
- c) $Q3 - Q2$
- d) $Q4 - Q1$

Answer: a) $Q3 - Q1$

71. What is the most common way to treat outliers in data?

- a) Remove them
- b) Replace them with mean or median values
- c) Use transformation techniques
- d) All of the above

Answer: d) All of the above

Inferential Statistics: Probability Distributions

72. Which of the following is a discrete probability distribution?

- a) Normal distribution
- b) Binomial distribution
- c) Exponential distribution
- d) Gamma distribution

Answer: b) Binomial distribution

73. The mean and standard deviation fully define which distribution?

- a) Normal distribution
- b) Poisson distribution
- c) Binomial distribution
- d) Uniform distribution

Answer: a) Normal distribution

Hypothesis Testing

74. The null hypothesis (H_0) typically states that:

- a) There is no significant effect or difference
- b) There is a significant effect or difference
- c) The test is inconclusive

d) The model is overfitting

Answer: a) There is no significant effect or difference

75. Which error occurs when the null hypothesis is incorrectly rejected?

a) Type I error

b) Type II error

c) Statistical error

d) None of the above

Answer: a) Type I error

76. A t-test is used when:

a) Sample size is small, and variance is unknown

b) Sample size is large and variance is known

c) Data is categorical

d) None of the above

Answer: a) Sample size is small and variance is unknown

ANOVA

77. What does ANOVA test for?

a) The difference between variances of groups

b) The difference between means of multiple groups

c) The correlation between variables

d) The normality of data

Answer: b) The difference between means of multiple groups

78. F-test is used to compare:

a) Two variances

b) Two means

c) Correlation coefficients

d) Skewness

Answer: a) Two variances

Preprocessing

79. What is data preprocessing?

a) Cleaning and transforming raw data into a usable format

- b) Building machine learning models
- c) Visualizing the data
- d) Performing feature selection

Answer: a) Cleaning and transforming raw data into a usable format

80. Which of the following is not a step in data preprocessing?

- a) Handling missing values
- b) Normalization
- c) Train-test split
- d) Encoding

Answer: c) Train-test split

Feature Selection (continued)

81. What is the purpose of feature selection?

- a) Reduce overfitting
- b) Improve model performance
- c) Decrease computation time
- d) All of the above

Answer: d) All of the above

82. Which method uses statistical tests to select features?

- a) SelectKBest
- b) PCA
- c) Normalization
- d) Bagging

Answer: a) SelectKBest

Introduction to Deep Learning

83. What type of data is typically processed by Convolutional Neural Networks (CNNs)?

- a) Text
- b) Images
- c) Numerical data
- d) Categorical data

Answer: b) Images

84. What is the function of dropout in a neural network?

- a) Normalize the data
- b) Prevent overfitting by randomly dropping neurons
- c) Reduce computation time
- d) Initialize weights

Answer: b) Prevent overfitting by randomly dropping neurons

Machine Learning Evaluation

85. What does the ROC curve represent?

- a) Accuracy vs Precision
- b) True Positive Rate vs False Positive Rate
- c) Precision vs Recall
- d) None of the above

Answer: b) True Positive Rate vs False Positive Rate

86. Which evaluation metric is sensitive to imbalanced datasets?

- a) Accuracy
- b) Precision
- c) Recall
- d) Both b and c

Answer: d) Both b and c

Clustering (continued)

87. What does the silhouette score measure?

- a) Variance within clusters
- b) Separation between clusters
- c) The distance between centroids
- d) None of the above

Answer: b) Separation between clusters

88. Which clustering method is density-based?

- a) KMeans
- b) DBSCAN
- c) Agglomerative Clustering
- d) PCA

Answer: b) DBSCAN

Artificial Intelligence

89. What is the goal of artificial intelligence?

- a) To build intelligent agents that can make decisions
- b) To perform complex computations
- c) To generate random data
- d) To visualize information

Answer: a) To build intelligent agents that can make decisions

90. Which of the following is an application of AI?

- a) Natural Language Processing (NLP)
- b) Image Recognition
- c) Autonomous Vehicles
- d) All of the above

Answer: d) All of the above

Neural Networks

91. What is forward propagation in a neural network?

- a) Calculating the output by passing data through the layers
- b) Optimizing weights
- c) Normalizing the data
- d) Scaling features

Answer: a) Calculating the output by passing data through the layers

92. What is the role of an optimizer in training a neural network?

- a) Initialize weights
- b) Minimize the loss function
- c) Add regularization
- d) Increase learning rate

Answer: b) Minimize the loss function

93. What is overfitting?

- a) A model performs well on training data but poorly on unseen data
- b) A model performs well on unseen data but poorly on training data
- c) A model that never converges

d) None of the above

Answer: a) A model performs well on training data but poorly on unseen data

94. Regularization helps to:

- a) Reduce overfitting
- b) Improve accuracy on training data
- c) Normalize data
- d) None of the above

Answer: a) Reduce overfitting

95. Which of the following is not a machine learning algorithm?

- a) Linear Regression
- b) Random Forest
- c) Monte Carlo Simulation
- d) Support Vector Machines

Answer: c) Monte Carlo Simulation

96. Which Python library is most commonly used for machine learning?

- a) NumPy
- b) Pandas
- c) Scikit-learn
- d) Matplotlib

Answer: c) Scikit-learn

97. The main goal of normalization is to:

- a) Scale data to fit within a specific range
- b) Reduce variance
- c) Improve accuracy
- d) None of the above

Answer: a) Scale data to fit within a specific range

98. What does an elbow method plot show?

- a) The optimal number of clusters
- b) The relationship between features
- c) Overfitting detection
- d) Variance reduction

Answer: a) The optimal number of clusters

99. Which metric is most suitable for binary classification?

- a) Mean Absolute Error
- b) F1 Score
- c) R-squared
- d) Mean Squared Error

Answer: b) F1 Score

100. Which Python library is best for deep learning?

- a) TensorFlow
- b) Scikit-learn
- c) Pandas
- d) NumPy

Answer: a) TensorFlow
